The Kidney

Exam code: 0610 & 0970

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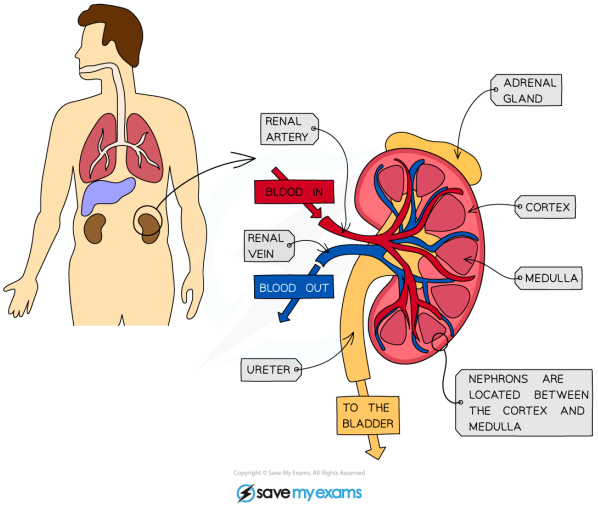
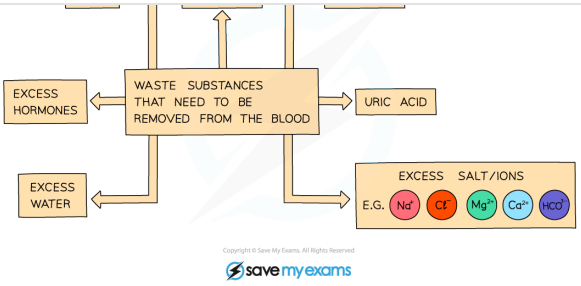
The Kidney & the Nephron: Extended

- The kidneys are located in the back of the abdomen and have two important functions in the body:
  - **They regulate the water content of the blood** (vital for maintaining blood pressure)
  - **They excrete the toxic waste products of metabolism** (such as urea) and substances in excess of requirements (such as salts)



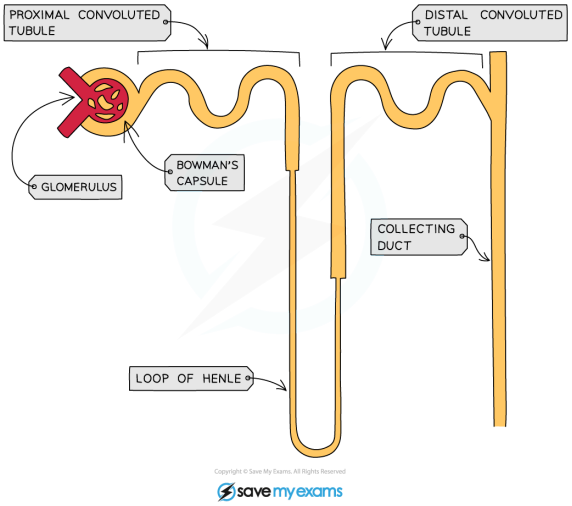
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



Waste substances

The Nephron

- Each kidney contains around a million tiny structures called **nephrons**, also known as **kidney tubules** or **renal tubules**
- The nephrons start in the **cortex** of the kidney, loop down into the **medulla** and back up to the cortex
- The contents of the nephrons drain into the innermost part of the kidney and the **urine collects** there before it flows into the **ureter** to be carried to the **bladder** for storage



Structure of a nephron

1) Ultrafiltration

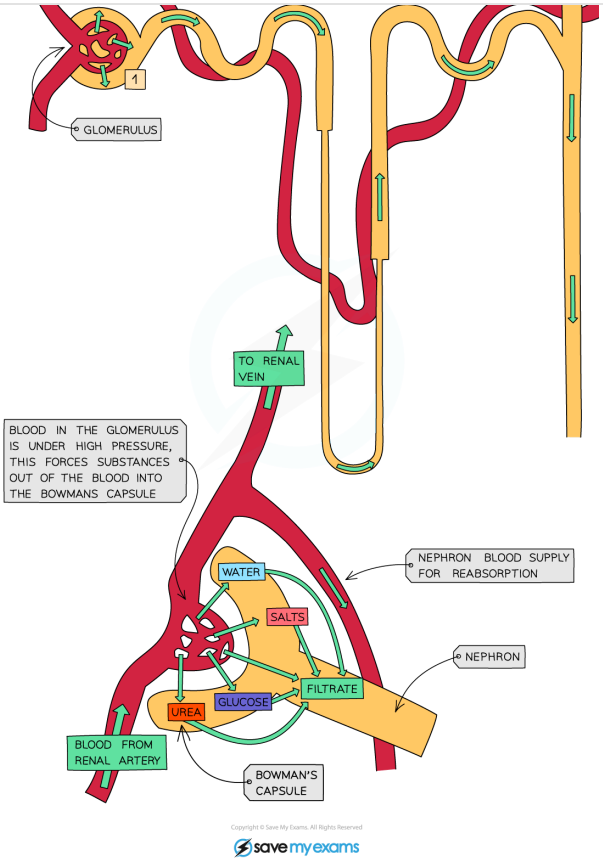


Diagram showing the process of ultrafiltration

- Arterioles branch off the **renal artery** and lead to each nephron, where they form a knot of capillaries (the **glomerulus**) sitting inside the cup-shaped **Bowman's capsule**
- The capillaries get **narrower** as they get further into the glomerulus which **increases the pressure** on the blood moving through them (which is already at high pressure because it is coming directly from the renal artery which is connected to the **aorta**)
- This eventually causes the smaller molecules being carried in the blood to be **forced out of the capillaries and into the Bowman's capsule**, where they form what is known as the filtrate
- This process is known as **ultrafiltration**
- The substances forced out of the capillaries are: **glucose, water, urea, salts**
- Some of these are useful and will be **reabsorbed back into the blood** further down the nephron

Components of filtrate:

WATER	LOOP OF HENLE AND COLLECTING DUCT
SALTS	LOOP OF HENLE
GLUCOSE	PROXIMAL (FIRST) CONVOLUTED
UREA	NOT REABSORBED

2) Selective Reabsorption

Reabsorption of Glucose

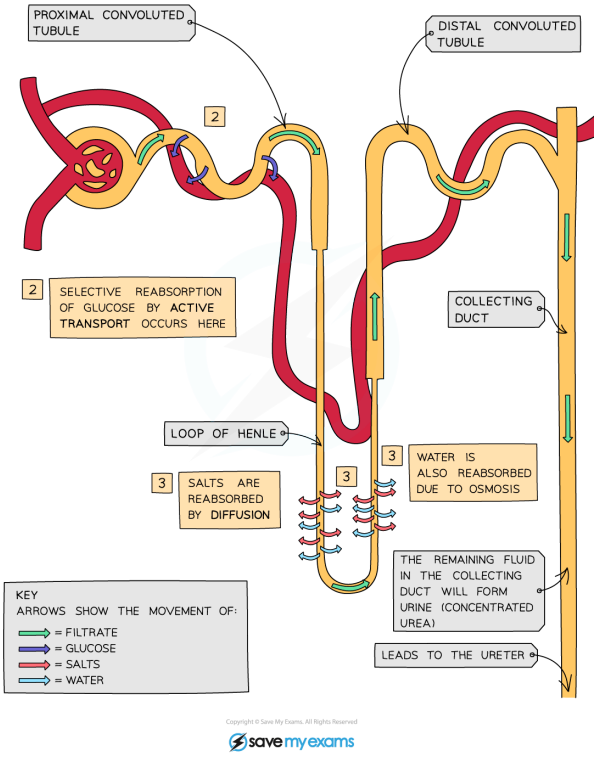


Diagram showing the reabsorption of glucose

- After the glomerular filtrate enters the Bowman's Capsule, **glucose** is the first substance to be reabsorbed at the **proximal (first) convoluted tubule**
- This takes place by **active transport**
- The nephron is adapted for this by having **many mitochondria** to provide energy for the active transport of glucose molecules
- Reabsorption of glucose **cannot take place anywhere else in the nephron** as the gates that facilitate the active transport of glucose are only found in the proximal convoluted tubule
- In a person with a normal blood glucose level, there are enough gates present to remove all of the glucose from the filtrate back into the blood
- People with **diabetes** cannot control their blood glucose levels and they are often very high, meaning that not all of

it continues in the filtrate and **ends up in the urine**
- This is why one of the first tests a doctor may do to check if someone is diabetic is to test their urine for the presence of glucose

Reabsorption of Water & Salts

- As the filtrate drips through the **Loop of Henle** necessary salts are reabsorbed back into the blood by **diffusion** and **active transport**
- As salts are reabsorbed back into the blood, **water** follows by **osmosis**
- Water is also reabsorbed from the collecting duct in different amounts depending on how much water the body needs at that time

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